

Hyesung Lee

Robotics · Physical AI · Dexterous Manipulation

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Introduction

I'm Hyesung Lee, a researcher deeply interested in Physical AI — particularly **Dexterous Manipulation** using multi-fingered robotic hands. My work focuses on enabling humanoid robots to manipulate objects by modeling force-level interactions through learning-based methods, going beyond purely kinematic approaches.

Research Interests:

- Dexterous Manipulation, Representation Learning for Visuo-Tactile & Kinesthetic Fusion, Vision-Haptic-Language-Action (VHLA) Models
- Reinforcement Learning, Imitation Learning
- Learning-based approaches for dexterous robot hand control

Education

Mar. 2026 – present	Ph.D. in Kim Jaechul Graduate School of AI, KAIST & KIST Advisor: Sungwook Yang (KIST) & Kimin Lee (KAIST)	Seoul, S. Korea
Feb. 2024 – Feb. 2026	M.S. in Computer Science, Korea University & KIST GPA: 4.31/4.50 Advisor: Sungwook Yang (KIST) & Jinkyu Kim (Korea Univ.) Thesis: <i>Learning to Refine Human Motion into Physically Feasible Robotic Actions for Dexterous Manipulation</i>	Seoul, S. Korea
Mar. 2017 – Aug. 2023	B.S. in School of Robotics, Kwangwoon University GPA: 3.76/4.50, Major GPA: 3.89/4.50 Advisor: Jin-Oh Kim Club: ROBIT (Robotics Competition Group)	Seoul, S. Korea

Work Experience

Intern — KIST (Korea Institute of Science and Technology) – Research on endomicroscopic image SLAM and sensor fusion – Performed photo-theranosis system integration	Mar 2023 – Aug 2024
Contract Worker — Kakao Mobility – Built a promotional autonomous driving mobile robot for the Kakao NEMO 2022 Conference – Researched stereo camera visual odometry	Dec 2021 – May 2022
Intern — Stryx (Startup Company) – Developed road data acquisition software using 3D LiDAR, GNSS, and machine vision camera – Developed a sensor calibration visualization tool using ROS Rviz	Sep 2021 – Dec 2021

International Publications

DexSynRefine: Synthesizing and Refining Human-Object Interaction Motion for Physically Feasible Dexterous Robot Actions Hyesung Lee, Hyunwoo Jung, Si-Hwan Heo, Sungwook Yang <i>arXiv Preprint (Under Review)</i> Project Page	2026.06
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UniDex-ViTac: Learning Unified Visuo-Tactile Dexterous Manipulation Policy from Human Video Data 2026.06
Hyesung Lee, Si-Hwan Heo, Sungwook Yang
IEEE ICRA 2026 (Beyond Teleoperation Workshop) — 🏆 **Best Paper Award** (\$500 Prize) | [Project Page](#)

Real-Time SLAM-Guided Closed-Loop Photodynamic Therapy with Pixel-Accurate Light-Dose Control 2026.01
Hyesung Lee, Sungwook Yang
IEEE RA-L / ICRA 2026

DexRefine: Refine Human Motion to Physically Feasible Robotic Actions 2025.10
Hyesung Lee, Si-Hwan Heo, Sungwook Yang
IEEE Humanoids 2025 (Dexterous Humanoid Manipulation Workshop)

Real-Time Endoscopic Image Mosaicking with an EKF-based Sensor Fusion Approach 2023.05
Kim Jin, Hyesung Lee, Sungwook Yang
IEEE EMBC 2023

Domestic Conference

Torque and Energy Regularization Penalty Effects in Reinforcement Learning Based Dexterous Hand Object Manipulation 2025.04
Hyesung Lee, Sungwook Yang
KSME Dynamics, Robot, Control Div. 2025

Deep Reinforcement Learning-Based Transformation of Human Motion Capture Data into Robot Actions to Reduce the Embodiment Gap between Humans and Robots 2025.02
Hyesung Lee, Sungwook Yang
KROS 2025

Real-Time Photo-Theragnosis of Brain Tumor Using Handheld Endoscopic Probe 2024.02
Hyesung Lee, Sungwook Yang
KROS 2024

Real-Time Photo-Theragnosis of Brain Tumor Using Handheld Endoscopic Probe 2023.11
Hyesung Lee, Sungwook Yang
KSME HQ 2023

Simultaneous Localization and Mapping of Endoscopic Images Using Optical Tracking System 2023.04
Hyesung Lee, Kim Jin, Sungwook Yang
KSME Bioengineering Div. 2023

Awards & Honors

2023.04	Paper Excellence Award — KSME Bio Engineering Div. Poster Presentation	KSME Bio Engineering Div.
2019.10	1st Place — National Defense Robot Contest	Hanwha Defense
2019.08	2nd Place — Busan Robot Contest, Turtlebot3 Autorace	Busan BEXCO
2018.11	3rd Place — 2018 R-BIZ Challenge, Turtlebot3 Autorace	Daegu EXCO
2018.10	5th Place — IRC 2018 Sumo Robot Competition	Ilisan KINTEX
2017.11	Grand Prize — Introduction to Engineering Design Exhibition	Kwangwoon Univ.

Projects

2019 Defense Robot Competition

May 2019 – Dec 2019

ROS, STM Cortex-M4, QT (Club ROBIT) — Controlled a BLDC motor using PID cascade control; developed a monitoring and pilot application using a Qt GUI over the UDP communication protocol.

Turtlebot3 Autorace Competition

Aug 2018, Jun 2019

ROS, 2D-LiDAR SLAM, Vision (Club ROBIT) — Lane detection using RANSAC; particle-filter-based localization; A* and potential fields for path planning and obstacle avoidance; YOLO-based sign detection.

2019 Hanyang Univ. Intelligent Model Car Competition

Jun 2019 – Aug 2019

Infineon TC237, Line Scan Camera (Club ROBIT) — Driving algorithm utilizing pure pursuit and PID control; developed a sensor monitoring and vehicle control program via Bluetooth connectivity with the Infineon TC237 board.

2022 Capstone Design

Sep 2022 – Dec 2022

ROS, 2D-LiDAR (Kwangwoon Univ. Capstone) — Applied Kalman filter and particle filter based tracking; adaptive-epsilon DBSCAN for object clustering; repulsive potential fields for obstacle avoidance; PI-controller-based driving algorithm.

Skills

Programming	C++ / C, Python, Matlab
Framework	Qt, TensorFlow, OpenCV, PyTorch, Hard Real-Time Framework
DevOps	ROS & ROS2, Git, Linux
Simulation	Isaac Lab & Isaac Sim
Languages	Korean, English